

A Search for Extended Circumgalactic Dust

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1 Motivation

- Motivate the research based on literature results: obviously Menard, but also Peek, and the theory papers. Do not include figures, except maybe in an appendix
- DO NOT CONGRATULATE

2 Methods

2.1 A Maximum Likelihood Estimator for Dust Extinction

- Should probably explain how A_v , reddening and R_v are related to one another
 - Linearly related! But dust-model dependent
 - We chose $R_v = 3.1$ by default, to allow an easy comparison to literature result, but others may be better...
- Develop our MLE formalism; reproduce equations
 - don't forget to include discussion of extinction vector (FM07 or whatever)
- Also want to include the ON and optimal vector generation, at least for a paragraph –YES
 - Think of this like a PCA; as long as the sum is non-negative, we're in good shape.
 - Dips positive or negative mean that the "principal component" under or overfits the actual trend at that distance
 - * little harder to connect to physical meaning for now – problem for theorists?
 - For development of "optimal" vector, show chi-squared plots, just to explain where we get these

- Discuss how and why we split A_V estimate into redshift bins
 - correlating *deviations from mean color space of background galaxies*
 - **HEY: did we deredden our magnitudes for galactic reddening? Are we hoping that we just "average" over dust? Because that's a bad idea**

2.2 Correlation Function Algorithm

- Treecorr algorithm, briefly (not our algorithm) but it is what we use. Correlating not gal-shear but gal-extinction TreeCorr¹ is a package for efficiently computing 2-point and 3-point correlation functions.
 - cite Jarvis, Bernstein, & Jain, 2004, MNRAS, 352, 338
 - Among others, the package returns the two-point number counts- κ correlation function, where κ can be any scalar quantities. To measure extinction, we substitute the value of the reddening vector for κ and use the position of the foreground sample as number counts
 - Mention spatial correlation as well: two-point correlation function of number counts (typically galaxy counts)
- Mention null tests: flexibility of algorithm allows us to easily correlate any two samples, so ideal for null tests like star-extinction

3 Datasets

- Background sample: redMaGiC
 - discuss algorithm, add some plots of color variance with MLE or color-color plots). Ref:
 - Like Peek catalog, these are also LRGs
 - * “an automated algorithm for selecting Luminous Red Galaxies (LRGs). The algorithm was specifically developed to minimize photometric redshift uncertainties in photometric large-scale structure studies. redMaGiC achieves this by self-training the color-cuts necessary to produce a luminosity-thresholded LRG sample of constant comoving density.”
 - area covers: $290^\circ \lesssim \alpha \lesssim 110^\circ$ of SDSS SV data

¹<https://github.com/rmjarvis/TreeCorr>

- very good photo-z algorithm: “redMaGiC catalog sampling the redshift range $z \in [0.2, 0.8]$. Our fiducial sample has a comoving space density of $10^{-3} \text{ (h}^{-1}\text{Mpc)}^{-3}$, and a median photoz bias ($z_{\text{spec}} - z_{\text{photo}}$) and scatter ($|z/(1+z)|$) of 0.005 and 0.017 respectively”
- SCOS as foreground:
 - “The archive also hosts the WISExSCOS Photometric Redshift catalogue, constructed by cross-matching ALLWISE and the SuperCOSMOS all-sky samples (All Sky Galaxy Catalogue) and employing the artificial neural network approach (the ANNz algorithm, Collister & Lahav 2004), trained on the GAMA-II redshift survey (Driver et al. 2011; Liske et al. 2015). The derived photometric redshifts have errors nearly independent of distance, with an overall accuracy of $\sigma_{\text{maz}}/(1+z) = 0.033$, and a very small percentage of outliers. These redshift estimates have a typical precision of 15%. The resulting WISExSuperCOSMOS sample contains (after appropriate masking) about 18.5 million galaxies with a median redshift of $z=0.2$.”
 - cite Bilicki et al. 2016, ApJS, 225, 1, 5.
 - Cuts applied were primarily redshift, also magnitude
- GALEX and (maybe) IIFSCz foregrounds
 - catalog used is Galex Ultraviolet Sources All-Sky Survey (GUVcat_AIS)
 - * cite Bianchi, Shiao, & Thilker (2017).
 - UV surveys are “unique resources for statistical studies of star-forming galaxies,” as young O&B stars emit heavily in the UV.
 - Selection: $nuv_mag < 0$ to ensure detection (?) and $largeobjsize < 66.0$ to eliminate objects whose large size would make probing at sub-arcminute scales futile.
 - * **side note: is that the reason we have that weird and high spike at low-r? That at 0.1 arcminutes, we’re just looking at the galaxies themselves?**
- Also SVM for DES foreground sample
 - The DES data we use comes from XXX
 - use Support Vector Machine (SVM) algorithm to generate a model, trained on COSMOS colors (very similar to DES colors) and use these to filter out galaxies at high-z
 - * perhaps move discussion of algorithm to 2
 - * ref needed

- Used Healpy to ensure that we are not correlating across empty space, which would have the effect of diluting the observed signal
 - Have some consistency plots showing overlap of two samples...

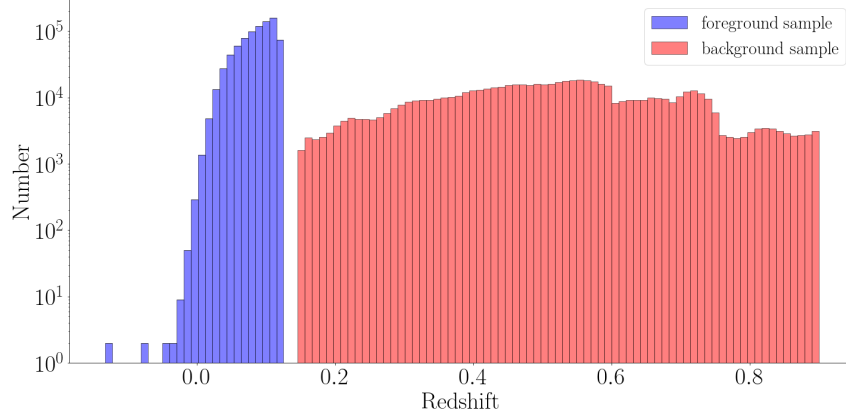


Figure 1: Redshift distribution of foreground WISExSCOS and background redMaGiC galaxy samples

4 Results

- Plot up all the results I do have... with dust vector, "optimal" vector for all samples
- MB show the "dust mass plot" but maybe also skip because it's not really a dust mass
- Star-bkg correlation is a good null test to show

5 Conclusions

- Start discussing consequences: here, compare to literature
- It might be cool to compare to MgII absorption plot, since that also shows evidence for extinction
 - Except that maybe that's controversial for actual MgII folks? Apparently they expect this to be anti-correlated
- For the literature figures, consider placing in appendix?